

Local knowledge for food security in Uganda; Postharvest practices in homegardens of the Ugandan Southwest

Cory Whitney

Faculty of Organic Agriculture, University of Kassel, Witzenhausen

Corresponding Author: acory.whitney@hsrw.eu

Introduction: Postharvest Losses (PHL) in East Africa destroy 20-60% of the food production (FAO 2011); in Uganda PHL are 35% for fruits and vegetables (Mulumba 2015), with up to 60% losses of more vulnerable crops such as tomatoes (Ogang 2011) primarily due to mold and rot (Kaaya et al. 2005; Kaaya & Kyamuhangire 2006). In seeking solutions to the overwhelming problems of food insecurity in East Africa, reducing PHL may be more effective than increasing yields or farming area (Lotze-Campen et al. 2008; Rockström et al. 2009) with a greater impact than improved cultivation methods (Goletti & Wolff 1999) to sustainably increase the volume of available food, without adverse effects on the regional ecological and economic situation (Hensel 2011). A study of botanical diversity and associated specific cultural practices of Ugandan homegardens in the Greater Bushenyi found many potential endogenous solutions to PHL. Methods Data collection took place around Rubirizi, Sheema, and Ishaka within the Greater Bushenyi Region of the southwest of Uganda, to assess plant use, and to understand related cultural and socio-economic factors. It sought to learn the plant diversity and associated traditional knowledge of farmers in 102 homegardens within 3 distinct geographic areas (forest edge, savannah, wetland). The study was guided, but not constrained (c.f. Vayda 1983), by the hypotheses that the three different biodiversity zones would contain distinct species and traditional knowledge. Villages were randomly selected from groupings of those that fell within the three pre-determined distinct biodiversity locations across the greater Bushenyi region. 11-12 households were randomly selected from each of 3 villages in the forest-edge communities of the far western highlands near Rutoto town in Rubirizi district (Kinoko-A, Kinoko-B, and Remitagu), 3 villages (cells) in the savannah and semi-urban areas of Bushenyi center around Ishaka town (Buhuma, Buramba, and Fort Jesus), and 3 villages in the wetlands of Sheema in the southeast near the town of Bugongi Kyarykunda, Rwabizi, and Nakashambya). Data collection followed a systematic approach fitting to the local culture and included three interviews, in February to April of 2015, based on semi-structured questionnaires about the uses for plants and postharvest practices. Cited species were recorded and identified by Ankole, Luganda, Lukiga, and Latin names with the help of farmers, local botanists, and botanical guidebooks. Specimens were conserved in a field press and are now part of the Makerere botany collection. All data was recorded and uploaded digitally in the field using Microsoft Excel (14.1.0, 2011). Names, uses, and postharvest treatments of recorded species were digitized and subsequently imported into the statistical package R (R version 3.1.1 Copyright (C) 2014 The R Foundation for Statistical Computing). Multiple linear regression analysis was performed using the “lm” function of the R package “psych”. Independent 2- group Mann-Whitney U Tests were performed using the R function “fisher.test” and “wilcox.test”. Results Local knowledge on postharvest practices for plants was a common factor to all gardens in the sample. By order of importance: farmers commonly processed millet (*Eleusine coracana*), dried and pounded to make a traditional sweet and sour fermented drink known as Bushera and a wet loaf known as Kalo; peanut (*Arachis hypogea*) was the next most important, sun-dried or roasted for cooking with meat or as a porridge; beans (*Phaseolus* spp.), peas (*Pisum sativum*), and soybeans (*Glycine max*), were commonly sundried and stored to be cooked as porridge or sold; cassava (*Manihot esculenta*), was often

